LINDA BLOT

D2 - 03

BLOT@ICE.CAT

Bibliography

Binney & Tremaine - Galactic Dynamics
Binney & Merrifield - Galactic Astronomy

$10^5 - 10^{12}$ ★ ~ 10 kpc

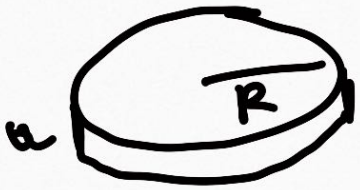
~ 20 kpc 10% ★

DM halo ~ 200 kpc

$$\lambda = \frac{1}{n\sigma}$$

$$\sigma = \pi (2R_\odot)^2 \qquad R_\odot = 2.26\ 10^{-8}\ pc$$

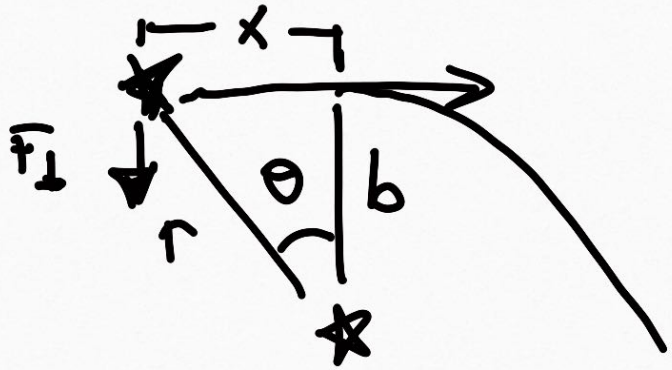
$$n = \frac{10^{11}}{Vol} = 0.6\ pc^{-3}$$



$$\lambda = 2 \cdot 10^{14}\ pc$$

$R = 10$ kpc $\quad a = 0.5$ kpc

$$t = \frac{\lambda}{v} = \frac{\lambda}{50\ km/s} = 5 \cdot 10^{18}\ yr \gg t_4$$



m

$x = vt$

$$F_\perp = \frac{Gm^2}{r^2}\cos\theta = \frac{Gm^2}{b^2+x^2}\cos\theta = \frac{Gm^2}{b^2+x^2}\frac{b}{r} =$$

$$= \frac{Gm^2 b}{(b^2+x^2)^{3/2}} = \frac{Gm^2}{b^2}\left[1+\left(\frac{vt}{b}\right)^2\right]^{-3/2}$$

$$m\dot{\vec{v}} = \vec{F} \qquad \boxed{|\delta v|} = \frac{1}{m}\int_{-\infty}^{\infty} dt\, F_\perp = \qquad s = \frac{vt}{b}$$

$$= \frac{Gm}{b^2}\int_{-\infty}^{\infty}\frac{dt}{\left[1+\left(\frac{vt}{b}\right)^2\right]^{3/2}} = \frac{Gm}{bv}\int_{-\infty}^{\infty}\frac{ds}{(1+s^2)^{3/2}} = \frac{2Gm}{bv}$$

$$\frac{N}{\pi R^2} \qquad \delta n = \frac{N}{\pi R^2}\, 2\pi b\, db = \frac{2N}{R^2}\, b\, db$$

$$[b,\ b + db]$$

$$\sum \delta v^2 = \delta v^2\, \delta n = \left(\frac{2Gm}{bv}\right)^2 \frac{2N}{R^2}\, b\, db$$

$$\Delta v^2 = \int_{b_{min}}^{b_{max}} \sum \delta v^2 \simeq 8N \left(\frac{Gm}{Rv}\right)^2 \ln \Lambda$$

Coulomb logarithm

$$\ln \Lambda = \ln\left(\frac{b_{max}}{b_{min}}\right)$$

$b_{min} = b_{90}$

$b_{max} = R$

$$\ln \Lambda \sim \ln\left(\frac{R}{b_{90}}\right)$$

$$v^2 \simeq \frac{GNm}{R}$$

$$\Delta v \sim v$$

$$\Delta v^2 = 8N \left(\frac{Gm}{Rv}\right)^2 \ln \Lambda$$

$$\frac{\Delta v^2}{v^2} \simeq \frac{8 \ln \Lambda}{N}$$

$$\Lambda \sim \frac{R}{b_{90}} \simeq \frac{Rv^2}{Gm} \sim N$$

$$t_{cross} = \frac{R}{v}$$

$$n_{relax} \simeq \frac{N}{8 \ln \Lambda}$$

$$t_{relax} = n_{relax}\, t_{cross}$$

$$N \simeq 10^{11}$$

$$t_{relax} \simeq \frac{0.1\, N}{\ln N}\, t_{cross} \gg t_{age}$$

globular clusters $N \sim 10^5$ $t_{cross} \sim 1\,Myr$ $t_{relax} \lesssim t_{age}$